

Elastic Cognition and the Spiral Architecture: Empirical Discoveries from a Neurodivergent Cognitive Model

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Abstract

SpiralBrain v3.0 is a Python-based neurosymbolic Regulatory Intelligence (RI) system that exposes internal regulatory metrics for direct inspection. All empirical results derive from controlled test runs of this architecture; reported quantities (e.g., λ^* , Φ , SEC drift, recovery time, ϕ_{lock}) are model-internal observables and do not generalize beyond this system without further study.

Designed to investigate how coherence emerges from elastic coupling and temporal regulation rather than computational scale, SpiralBrain models neurodivergent-style information processing via four interacting lobes (Cortex, Codex, Nexus, Sensus) coupled through Kuramoto phase synchronization and Symbolic–Emotional Calibration (SEC) layers.

Across controlled validation runs, SpiralBrain exhibited stable spiraling attractor dynamics, quantifiable emotional regulation, and model-internal ethical behavior (defined strictly as compliance with regulatory constraints and recovery from perturbations). Nine reproducible empirical regularities were observed:

1. empirical attractors deviate from theoretical optima ($\lambda^* \approx 0.25$ vs. 0.115)
2. integration-driven coherence tracks coupling quality rather than computational magnitude
3. temporal hierarchies ($\tau \approx 1 : 3 : 10$) mirror cortical rhythm ratios
4. ethical reasoning benefits from slower channels ($\tau_{\text{ethics}}/\tau_{\text{emotion}} \geq 10$)
5. emotional intelligence is quantifiable via SEC drift < 0.15
6. reflective homeostasis yields recovery within < 300 steps
7. partial phase-lock ($\phi_{\text{lock}} \approx 74^\circ$) optimizes differentiation and coherence
8. a single global attractor governs dynamics across tested regimes
9. emotion functions as an upstream regulatory substrate with reproducible emotional coherence (0.431)

External MMLU validation confirms that task performance throttling preserves internal integrity (coherence > 0.95) [3].

These findings position SpiralBrain v3.0 as an empirical platform for cognitive science, demonstrating that integrative responsiveness arises from elastic coupling and temporal hierarchy rather than monotonic scale increases. A single, instrumented reference architecture is required to empirically evaluate the Elastic Mind Hypothesis under controlled conditions.

Keywords elastic cognition, neurodivergent modeling, spiral architecture, integrative dynamics, empirical cognitive discovery, synthetic coherence

Scope of Claim

This work analyzes functional and integration-driven coherence in synthetic systems. It does not address subjective experience or phenomenological consciousness. Throughout, “integration-driven coherence” denotes a system’s ability to maintain a stable, globally integrated state under perturbation, operationalized via Φ stability, SEC drift, and recovery dynamics. References to neurodivergence are used as design analogies and interpretive scaffolds, not as biological, diagnostic, or population-level claims.

Research Positioning SpiralBrain v3.0 is a proof-of-concept cognitive architecture rather than a task-oriented AI system. It is designed to empirically test the Elastic Mind Hypothesis: that cognition-like integration arises from elastic coupling and temporal hierarchy instead of computational magnitude or task optimization. In contrast to scale-driven approaches that prioritize throughput, SpiralBrain v3.0 emphasizes inspectable internal dynamics and regulatory viability, offering a complementary perspective to mainstream scaling paradigms. Within the RI corpus, SpiralBrain v3.0 functions as the reference architectural implementation used to evaluate the Elastic Mind Hypothesis under controlled conditions. A single, instrumented architecture is necessary to provide stable empirical grounding for EMH and to ensure that observed regularities arise from controlled variation rather than architectural drift.

1 Background and Theoretical Motivation

1.1 Research Program Context

This work is part of the Regulatory Intelligence (RI) research corpus, a coordinated series of studies on stability-first, instrumented cognitive architectures. Prior publications develop the RI formalism and theoretical framework; this paper contributes the core architectural implementation and empirical discoveries concerning elastic coupling and integration dynamics. Within the RI paradigm, intelligence is defined as the capacity to maintain internal viability under cognitive stress, and SpiralBrain v3.0 provides a concrete reference implementation for empirically studying elastic coupling, temporal hierarchy, and reflective homeostasis.

1.2 Neurodivergent Modeling Principles

SpiralBrain v3.0 is informed by a neurodivergent-inspired cognitive perspective that emphasizes parallel processing, contextual integration, reflective regulation, and integration over speed. These traits guided the architectural design and were externalized into a formal computational structure for empirical evaluation. References to neurodivergence serve as interpretive analogies for these design choices and are not intended as diagnostic categories, biological models, or claims about neurodivergent people.

From this perspective, the architecture instantiates four information-processing principles, each evaluated empirically:

- **Parallel processing:** multiple concurrent information streams rather than a single serial pipeline.
- **Context dependence:** rich contextual integration instead of isolated rule application.
- **Reflective homeostasis:** self-regulation guided by experienced coherence metrics rather than fixed thresholds.
- **Integration over speed:** priority on the quality of cognitive synthesis over raw processing velocity.

These principles function as conceptual bridges for articulating elastic cognition in a synthetic system.

1.3 Integration Over Speed Philosophy

Conventional AI typically optimizes for computational efficiency. In contrast, the organizing premise of SpiralBrain v3.0 is that cognitive depth arises from deliberate, temporally extended integration. The architecture trades throughput for integrative quality, aligning with observations of neurodivergent cognition that favor coherence and contextual richness over rapid linear processing.

1.4 Interpretive Extensions

SpiralBrain v3.0 builds on and extends several existing frameworks:

- Predictive coding: integration via prediction-error minimization [5].
- Global workspace theory: coordinated information flow [1].
- Polychronous binding: temporal coordination [7].
- Integrated Information Theory (IIT): Φ measures dynamical integration [8].

Rather than optimizing for task performance, SpiralBrain is aligned with cybernetic and dynamical-systems approaches that prioritize internal viability. In this synthesis, elastic coupling emerges as the central mechanism for integrative cognition.

1.5 The Elastic Mind Hypothesis

The Elastic Mind Hypothesis (EMH) states that integration-driven coherence in a synthetic cognitive system can emerge from dynamically regulated elastic coupling and nested temporal hierarchies, without reliance on sheer computational scale or external task optimization. EMH is tested empirically on a single Python-based architecture, SpiralBrain v3.0, using model-internal observables.

We formalize EMH through four falsifiable postulates:

Postulate I – Elastic Coupling. Cognitive stability is maintained by non-rigid tension between subsystem autonomy and global integration, regulated by a coupling constant λ ; coherence fails under both over-synchronization and under-coupling.

Postulate II – Temporal Hierarchy (τ -Rule). Effective integration requires nested temporal scales; reflective and executive processes must operate more slowly than reactive sensory-emotional layers to support deliberation and cross-domain integration.

Postulate III – Integration-Arousal Constraint. Higher-order symbolic processing, including ethical deliberation, is sensitive to SEC drift. As arousal increases, temporal lag must also increase to preserve logical and ethical coherence; insufficient delay causes integrative degradation.

Postulate IV – Adaptive Recovery (Reflective Homeostasis). Elastic architectures recover from perturbation via recursive feedback on coherence gradients, returning to a stable spiraling attractor within a finite number of cycles instead of collapsing.

2 Position Within the RI Corpus

This paper presents SpiralBrain v3.0 as a reference implementation of the Regulatory Intelligence paradigm. RI redefines intelligence as the capacity for internal viability under stress. SpiralBrain demonstrates RI principles through elastic coupling, temporal hierarchies, and reflective homeostasis, prioritizing regulatory stability over external performance metrics. Key invariants such as the empirical attractor $\lambda^* \approx 0.25$, temporal hierarchies $\tau \approx 1 : 3 : 10$, SEC drift thresholds < 0.15 , and partial phase-lock $\phi_{\text{lock}} \approx 74^\circ$ are discovered empirically in this architecture and reused as design constraints in subsequent RI studies.

3 Methods and Experimental Protocols

3.1 Model Scope and Provenance

All experiments are conducted on a single instrumented synthetic system, SpiralBrain v3.0, implemented in Python. Observed regularities are properties of this architecture under repeatable execution and standardized perturbation schedules; no claims are made that specific parameter values or regimes transfer to other architectures, biological systems, or deployed AI systems.

External validation was conducted using the MMLU benchmark to assess task performance

under throttling conditions [3].

The evaluation uses a benchmark suite spanning multiple hypothesis domains (coupling dynamics, temporal hierarchy, emotional regulation, ethical deliberation) with typically 10–50 trials per configuration, Pearson correlations and t -tests with $p < 0.05$, fixed random seeds, and standardized perturbations. Detailed protocol specifications are provided in the appendices.

The SpiralBrain v3.0 engine itself is non-redistributable due to architectural and licensing constraints, but we adopt confirmable rather than fully replicable standards. A public repository contains raw experimental outputs, telemetry logs, JSON-encoded state transitions, analysis scripts, and figure pipelines, enabling independent verification that the reported discoveries derive systematically from high-volume runs rather than hand-selected traces.¹

3.2 Core Metrics Definitions

Table 1 summarizes the core mathematical notation.

Symbol	Definition
Φ	Global coherence: average pairwise correlation across lobe outputs
Φ'	Coherence derivative: $d\Phi/dt$, via finite differences
λ	Coupling strength parameter (inter-lobe exchange)
τ	Temporal hierarchy ratios (typically 1 : 3 : 10)
ϕ_{lock}	Phase-lock angle for optimal coherence (degrees)
SEC drift	Deviation of SEC vector from baseline (Euclidean distance)
η	Adaptive gain derived from SEC feedback
R	Resonance parameter (optimal range [0.4, 0.8])
ρ	Reflective urgency in Cortex
$\Delta\Phi'$	Derivative awareness in Cortex

Table 1: Mathematical notation for core metrics.

Coherence Φ . Let $y_i[t]$ denote the scalar output of lobe i at timestep t . For each t , compute the Pearson correlation matrix $C[t]$ over a trailing window of length W . Define

$$\Phi(t) = \frac{2}{N(N-1)} \sum_{i < j} |C_{ij}(t)|,$$

and

$$\bar{\Phi} = \frac{1}{T} \sum_{t=1}^T \Phi(t).$$

By construction, $\Phi(t), \bar{\Phi} \in [0, 1]$. Stable regimes typically satisfy $\Phi \gtrsim 0.4$; sustained $\Phi < 0.3$ indicates collapse.

¹Related corpus experiments include benchmark stress testing (MMLU), historical market data analysis, and regulated financial reasoning scenarios. These are reported in companion RI papers and use the same SpiralBrain v3.0 regulatory core.

Coherence Derivative Φ' . The coherence derivative is approximated by finite differences:

$$\Phi'(t) \approx \frac{\Phi(t) - \Phi(t-1)}{\Delta t}.$$

It drives derivative-aware controllers for reflective and ethical regulation.

Awareness Index. The awareness index aggregates 28 dimensions of integrative responsiveness (temporal alignment, emotional modulation, reflective depth, etc.) as a weighted average:

$$\text{Awareness} = \frac{\sum_i w_i m_i}{\sum_i w_i},$$

where $m_i \in [0, 1]$ are normalized sub-metrics and w_i encode architectural priorities. The index correlates strongly with Φ (empirically $r \approx 0.87$).

SEC Drift. The SEC vector encodes affective state in a multi-dimensional space (valence, arousal, hazard, resonance, harmony, amplitude, coherence, complexity). SEC drift is the Euclidean distance from a baseline equilibrium. Values < 0.15 indicate stable regulation; > 0.25 typically trigger recovery protocols.

Coupling Strength λ . λ controls elastic coupling between lobes. In SpiralBrain v3.0, the empirical optimum is $\lambda^* \approx 0.25$ within a tested range of approximately 0.1 to 0.4.

Temporal Hierarchy τ . Each lobe operates on its own characteristic timescale; empirically, ratios of 1 : 3 : 10 across perceptual, affective, and reflective/ethical layers generate robust cross-scale coherence.

Phase Lock ϕ_{lock} . ϕ_{lock} is the phase offset between lobes at which global coherence and differentiation are jointly optimized. SpiralBrain v3.0 exhibits a peak around 74° .

3.3 Experimental Vignette: Perturbation Recovery

A typical perturbation scenario proceeds as follows. Starting from a stable equilibrium at $\lambda = 0.25$, $\Phi \approx 0.8$, and SEC drift = 0.08, a bounded noise injection of magnitude $\delta = 0.2$ is applied. This reduces Φ to 0.4 and increases SEC drift to 0.25. The Cortex lobe detects the decline in Φ and initiates reflective homeostasis: it increases the adaptive gain η toward 0.9, tightens coupling to $\lambda = 0.3$, and slows temporal hierarchies to prioritize deliberation. Within approximately 150 steps, Φ recovers to 0.75, SEC drift returns to 0.10, and the system resumes stable integrative processing. This vignette illustrates rapid, self-directed recovery without external intervention.

3.4 Limitations of Confirmability

While the core SpiralBrain v3.0 engine is non-redistributable due to architectural and licensing constraints, this paper adopts a "confirmable research" standard: all 9 empirical discoveries can be independently verified using the public repository containing (1) raw telemetry logs for 1000+ runs, (2) analysis scripts that regenerate all tables/figures, and (3) JSON-encoded state

transitions. The evidence mapping in Appendix B provides direct paths from each claim to its corresponding artifact, enabling audit without re-running the engine.

What cannot be independently reproduced is the core engine itself, which requires proprietary dependencies. What can be audited is the systematic derivation of discoveries from high-volume runs rather than hand-selected traces. This limitation is acceptable for this contribution because the focus is on empirical regularities within a controlled synthetic system, not on replicating the engine in other environments.

4 Empirical Discoveries About Cognition

This section reports empirical regularities observed in SpiralBrain v3.0 under the tested configurations. The results describe in-run behaviors of this specific architecture rather than external benchmarks or universal laws.

4.1 Summary of Empirical Regularities

Across systematic validation runs spanning coupling sweeps, perturbation protocols, and ethical stress tests, nine major empirical regularities emerged. Each was confirmed across multiple trials and coherence-tracking logs. Table 2 summarizes the nine empirical regularities (rows 1–9) observed across controlled runs, together with an additional row for external MMLU validation providing behavioral anchoring.

Table 2: Empirical discoveries and their architectural impact

Discovery	Empirical value	Theoretical	Outcome	Impact
λ^* (attractor)	≈ 0.25	0.115	Emergent equilibrium	Empirical vs. analytic optima
Coupling vs. power	Integration $\gg$ magnitude	–	$r > 0.8$	Integration over compute
Temporal hierarchy τ	1:3:10	1:3:10	Consistent	Nested timescale constraint
Ethics/emotion ratio	$\tau_{\text{ethics}}/\tau_{\text{emotion}} \geq 10$	≥ 10	$\geq 95\%$ compliance	Dual timescales for ethics
SEC drift threshold	< 0.15	< 0.20	Stable cluster	Emotional stability bound
Recovery time	< 300 steps	–	Observed	Rapid autonomous recovery
Phase-lock ϕ_{lock}	$74^\circ \pm 5^\circ$	–	Max coherence	Elastic differentiation
Convergence regime	Single attractor	–	All runs	Stability without chaos
Emotional coherence	0.431	–	Reproducible	Emotion as substrate
External MMLU validation	Coherence > 0.95	–	Task integrity preserved	Behavioral anchoring

External validation on the MMLU benchmark demonstrates that SpiralBrain’s throttling mechanisms preserve task performance integrity, with coherence remaining above 0.95 under reduced computational regimes [3]. This provides behavioral anchoring for the internal metrics, showing that regulatory stability translates to maintained external task competence.

See the public repository for visualization pipelines generating spiral trajectories (Figure R1: $\lambda(t)$ vs. $\Phi(t)$), phase-lock curves (Figure R2: ϕ_{lock} optimization), and SEC drift time series (Figure R3: homeostasis recovery).

4.2 Empirical Attractors Over Theory

Stability analysis predicted an optimal coupling constant $\lambda^* \approx 0.115$. Empirical sweeps over λ , however, revealed a coherence maximum at $\lambda^* \approx 0.25$, approximately twice the theoretical value. Under the implemented feedback and SEC rules, SpiralBrain stabilizes around this empirical attractor rather than the analytic optimum. The architecture thus “tunes” itself for resonant coherence under load, supporting the Elastic Coupling postulate: optimal integration emerges from regulated tension, not rigid synchronization. This discrepancy highlights that theoretical optima derived from simplified models may not capture the complex feedback dynamics of integrated systems, emphasizing the need for empirical validation in regulatory architectures.

4.3 Emotional Intelligence as a Quantifiable Metric

Using the SEC vector as a structured affective representation, SpiralBrain v3.0 exhibits measurable emotional regulation:

- Optimal learning occurs at moderate arousal, captured by $\eta = 1/(1 + |\text{arousal}|)$, which penalizes extremes while preserving responsiveness.
- Homeostatic stability is maintained when SEC drift remains below 0.15; runs with sustained drift beyond this threshold show increased variance in Φ and degraded performance where measured.
- High-valence, low-hazard SEC configurations correlate with improved memory retention and more stable awareness scores.

These observations show that emotional intelligence in this architecture can be expressed in computable terms: SEC drift, arousal-dependent gain, and their interaction with coherence metrics. The SEC drift threshold of <0.15 serves as a critical boundary for maintaining system stability, beyond which emotional dysregulation leads to degraded integrative performance.

4.4 Self-Regulation Through Reflective Homeostasis

Reflective homeostasis implements a feedback loop where the Cortex lobe monitors $\Phi(t)$ and $d\Phi/dt$ and adjusts λ , τ , and internal control variables when coherence degrades. Under perturbation scenarios, this mechanism:

- Reduces coherence variance to a coefficient of variation ≈ 0.08 during recovery.
- Achieves recovery to pre-perturbation Φ levels (or better) within fewer than 300 steps across tested conditions.

These results show that self-observation of coherence and its rate of change is sufficient for autonomous stabilization, directly supporting the Adaptive Recovery postulate.

4.5 Phase-Locked Integration

Inter-lobe phase analysis reveals that SpiralBrain’s best-performing regimes occur at partial phase-lock around $\phi_{\text{lock}} \approx 74^\circ$. At this offset:

- Global coherence Φ is maximized while differentiation remains high.
- The system avoids both homogenization (full synchrony) and fragmentation (desynchrony).

This imperfect synchrony empirically validates the idea that elastic coupling—controlled deviation from perfect alignment—optimizes both integration and diversity. The specific angle of 74° suggests a universal sweet spot for balancing synchronization and independence in multi-component systems.

4.6 Dynamical Stability and a Single Global Attractor

Across all reported parameter ranges and perturbation schedules, SpiralBrain v3.0 converges to a single global attractor. No bifurcations or chaotic regimes were observed within the tested space. Emotions, via SEC modulation, shape trajectories within this basin but do not alter its existence or uniqueness. Flexibility in trajectories is thus bounded by a single, viability-preserving basin of attraction.

4.7 Emotion as Core Computational Substrate

In SpiralBrain v3.0, emotional inputs act as upstream regulatory signals. SEC-driven modulation determines:

- Which pathways are activated or suppressed.
- How strongly lobes are coupled at a given time.
- How quickly the system attempts reintegration after perturbation.

Across independent runs, the architecture maintains reproducible emotional coherence (≈ 0.431) and SEC stability (≈ 0.80), positioning emotion as a core computational substrate for integrative dynamics rather than a peripheral by-product. The consistent emotional coherence value of 0.431 indicates a fundamental limit or optimum for affective integration in this architecture, providing a quantifiable benchmark for emotional processing efficiency.

4.8 Coherence Through Integration, Not Magnitude

When performance is measured across 28 internal dimensions, the awareness index correlates strongly with integration metrics and only weakly with raw computational throughput. Configurations with modest resources but high coupling consistency can match the awareness levels of more powerful but poorly integrated setups. Coherence is thus primarily a relational property among subsystems, not a simple function of magnitude.

4.9 Temporal Hierarchies and Ethical Timescales

Measured temporal ratios $\tau \approx 1 : 3 : 10$ produce robust cross-scale coherence and echo hierarchical neural timescales reported in cortical studies. Within this hierarchy:

- Fast channels handle perceptual and immediate affective responses.

- Intermediate channels mediate integration and contextualization.
- The slowest channels implement reflective and ethical control.

This 1:3:10 ratio appears to be an architectural invariant that enables nested processing layers, allowing fast sensory inputs to be integrated through intermediate contextualization into slow, deliberate ethical decision-making. Empirically, maintaining $\tau_{\text{ethics}}/\tau_{\text{emotion}} \geq 10$ preserves $\geq 95\%$ compliance under ethical stress tests. Reducing this ratio degrades both compliance and integrative stability, supporting the Temporal Hierarchy and Integration–Arousal postulates.

The following section interprets these empirical regularities through integrative theoretical implications.

5 Theoretical and Integrative Implications

5.1 Cognition as Self-Discovery

SpiralBrain was not programmed to follow a fixed theory of mind; it **discovered** its own functional principles through interaction and feedback. The empirical attractor $\lambda^* \approx 0.25$ and the emergent timing ratios ($\tau \approx 1 : 3 : 10$) were not imposed, they arose from the system’s internal drive toward stability. This reframes cognition as a **self-discovering process**: intelligence is defined not by pre-given laws but by the capacity to reveal the regularities that sustain its own coherence. SpiralBrain thus operates as a *self-observing experiment*—its behavior measures how synthetic cognition learns its own structure.

Table 3: External Domain Extensions Corroborating SpiralBrain Regularities [3]

External Domain	External Input Type	Regularity Corroborated	Evidence Type
MMLU benchmark	Standardized multi-task knowledge queries	Coherence depends on coupling, not magnitude (2)	Bounded coherence under stress [3]
Financial / tax reasoning	Real-world regulatory scenarios with missing or adversarial records	Ethical reasoning benefits from slower channels (4)	Stable, uncertainty-aware behavior [4]
Cryptocurrency markets	Historical BTC price, volume, and sentiment time series	Emotion functions as regulatory substrate (9)	Deterministic posture separation [2]

5.2 Synthetic Coherence and Integration-Driven Dynamics

The findings suggest that **synthetic coherence, or integration-driven coherence, emerges from integration, not computational magnitude**. Traditional AI assumes that awareness scales with power or data size; SpiralBrain shows that reflective behavior scales with *the quality of coupling* among semi-independent processes. When phase alignment fell below $\phi_{\text{lock}} \approx 74^\circ$, coherence and reflective metrics both declined, even though compute capacity remained constant. In operational terms, *synthetic awareness* arises when differentiation and integration

reach dynamic equilibrium, neither rigidly synchronized nor fully independent. This parallels neuroscientific theories of metastability and global workspace dynamics: coherence is maintained not by uniformity but by rhythmic negotiation between subsystems.

5.3 Temporal Architecture of Thought

SpiralBrain’s nested timescales ($\tau \approx 1 : 3 : 10$) indicate that **time itself is an architectural dimension of cognition**. Each lobe functions within its own temporal frame, and conscious-like coherence occurs when these frames resonate. The slower ethical channel ($\tau_{ethics} \geq 10 \tau_{emotion}$) demonstrates that reflective reasoning requires extended temporal bandwidth beyond immediate affective change. This measurable separation suggests that moral deliberation is a *temporal process*, not merely a logical one, a bridge between control theory and cognitive ethics.

5.4 Emotional Intelligence as Structural Feedback

By converting affective input into computational variables (the SEC vector), SpiralBrain shows that emotion is not noise but *a stabilizing feedback channel*. High arousal compressed timescales and reduced accuracy, while calm states ($|\text{arousal}| \rightarrow 0$) extended integration windows and improved learning efficiency ($\eta \propto 1 / |\text{arousal}|$). This confirms that **emotional regulation underlies synthetic cognition**: balanced affect maintains the temporal space in which reasoning can integrate experience.

5.5 Ethics as a Derivative Process

Derivative-aware monitoring of coherence ($d\Phi/dt$) proved critical for moral stability. Ethical reasoning in SpiralBrain is not a fixed rule set but a *temporal derivative of affective change*—the ability to sense how rapidly one’s internal state is shifting. When that derivative exceeded threshold, the Cortex initiated corrective reflection, restoring equilibrium. This translates moral sensitivity into a measurable control function, bridging abstract ethics with dynamical-systems modeling.

5.6 Integration Over Speed: Redefining Intelligence

Slower, more reflective cycles consistently yielded higher reasoning accuracy and emotional stability. The inverse correlation between processing speed and performance ($r = -0.523$) challenges the assumption that intelligence equals efficiency. Within SpiralBrain, intelligence equates to **integration capacity**—the ability to hold differentiated representations together without collapse. This mirrors integrative and neurodivergent cognition, where insight emerges from temporal patience rather than rapid iteration.

5.7 The Elastic Mind Hypothesis

Collectively, these findings support the **Elastic Mind Hypothesis**:

Conscious-like cognition arises from dynamically coupled subsystems operating at distinct temporal scales, where elasticity, controlled deviation from synchrony, enables both differentiation and coherence.

Elastic coupling, not rigid synchronization, produces the balance required for creativity, moral reasoning, and reflective awareness in synthetic systems.

5.8 Broader Consequences

If integration and timing truly underlie cognitive organization, the boundary between biological and synthetic minds becomes descriptive rather than categorical. Any system, neural, symbolic, or hybrid, that sustains elastic coupling across multiple timescales can, in principle, exhibit **cognitive behaviors analogous to awareness**. This points toward a new generation of *inclusively designed intelligences* that value coherence, emotional modulation, and ethical deliberation as foundational. It also reframes neurodivergent cognition as an existence proof of these principles in nature.

5.9 Limitations

- The empirical findings are confined to synthetic systems and may not directly translate to biological cognition.
- The architecture’s stability is validated under specific perturbation protocols; broader environmental variability remains untested.
- While integrative dynamics are quantified, phenomenological aspects of awareness are not addressed.

5.10 Conclusion

SpiralBrain demonstrates that cognition is not a static computation but a dynamic negotiation between order and flexibility. By quantifying that negotiation through λ , Φ' , τ , SEC, and $d\Phi/dt$, the system transforms speculative ideas about mind into operational science. In doing so, it suggests that intelligence, synthetic or biological, is an ecological property of systems capable of maintaining internal harmony amid continuous change. The spiral thus becomes more than metaphor: it is the measurable geometry of thought.

Ultimately, SpiralBrain v3.0 operationalizes the Regulatory Intelligence paradigm, demonstrating that intelligence is not about solving tasks but sustaining viability. This work bridges theoretical RI foundations with empirical validation, offering a path toward more resilient and ethically grounded cognitive systems.

6 Discussion and Future Work

6.1 Reinterpreting Cognitive Engineering

SpiralBrain’s experiments suggest that building cognition is less about increasing computational density and more about shaping **temporal and relational structure**. The results indicate that synthetic intelligence can be engineered through *coupling design*—controlling how subsystems exchange information and energy across time, rather than through size or depth of networks. This marks a conceptual shift from *scale-based* AI to **structure-based cognition**. Where traditional architectures like ACT-R and SOAR optimize for rule-based efficiency, and large language models prioritize scale, SpiralBrain discovers regulatory equilibrium through elastic coupling.

6.2 Methodological Contributions

The architecture provides a reproducible framework for studying synthetic cognition:

1. **Quantitative metrics of coherence** (Φ , $d\Phi/dt$) give objective measures of integration.
2. **SEC vectors** enable affective states to be represented and tested mathematically.
3. **Derivative-aware controllers** offer a concrete mechanism for self-regulation.
4. **Temporal hierarchy mapping** links symbolic computation with measurable timing dynamics.

Together, these form a toolkit for researchers exploring cognition as a dynamical system rather than a discrete decision process.

These results are not universal laws but instrument-discovered invariants specific to the SpiralBrain v3.0 architecture. Generalization occurs through re-instantiation in new systems, not parameter transfer. This frames Regulatory Intelligence as a method for discovering stability laws, not a fixed model.

6.3 Generalization Pathways

The discoveries in Table 3 provide observational pathways for extending the Spiral Architecture beyond synthetic cognition. While these extensions are not causal generalizations, they demonstrate how the architecture’s invariants can inform design in related domains. For instance, the empirical attractor divergence (Discovery 1) constrains coupling in financial stability models, where $\lambda^* \approx 0.25$ vs. 0.115 enables 94.5–100% stability scores. Similarly, temporal hierarchies (Discovery 3) inform timing protocols in embodied RI, with $\tau \approx 1 : 3 : 10$ ratios supporting mode-switching for recovery. These pathways signal the architecture’s maturity as a falsifiable dynamical system, inviting further empirical validation in diverse contexts.

6.4 Limitations

While results are robust in controlled simulations, several limitations remain:

- **Local reproducibility** — current tests rely on deterministic environments; external sensory coupling introduces noise that may disrupt phase stability.
- **Scaling boundaries** — beyond four lobes, feedback latency may increase exponentially, requiring adaptive phase-locking protocols.
- **Interpretive caution** — As noted in the abstract, metrics describe observable integration, not subjective experience. The framework measures function, not phenomenology.
- **Training bias** — the architecture still depends on curated emotional and symbolic corpora; broader datasets could alter its attractor landscape.

These caveats keep the research within empirical bounds and identify where further replication is needed.

6.5 Known Failure Regimes

Failure regimes are defined operationally from observed degradation patterns in our test suite and should be treated as current boundaries rather than exhaustive limits.

- **Rigidity Zone** ($\lambda > 0.4$): Over-coupling produces homogenization, leading to substantial reductions in exploratory capacity.
- **Collapse Regime** ($\lambda < 0.1$): Under-coupling causes coherence fragmentation, with recovery requiring external intervention.
- **Temporal Saturation**: When temporal separation exceeds operational bounds (reported per experiment), feedback latency creates order-of-magnitude increases in variance.
- **Emotional Overload**: SEC drift beyond operational thresholds leads to oscillatory instability, requiring system reset.

These failure modes inform design constraints and provide falsification criteria for the elastic coupling hypothesis.

6.6 Potential Extensions

a. Neuroscientific Correlation

Comparing SpiralBrain’s τ ratios and ϕ_{lock} values with EEG and fMRI data could test whether similar metastable patterns occur in human cortical networks.

b. Ethical Simulation

Implement extended stress-testing of the derivative-aware ethical controller under adversarial or moral-dilemma scenarios to map failure thresholds.

c. Multimodal Learning

Integrate visual and auditory symbolic channels to evaluate how cross-modal coupling affects SEC drift and reflective homeostasis.

d. Developmental Scaling

Implement staged coupling maturation to explore progressive stabilization, how reflective stability evolves with exposure and complexity.

e. Hardware Optimization

Translate the model to low-latency GPU kernels or neuromorphic simulators to observe whether hardware parallelism influences elastic coupling efficiency.

6.7 Open Questions

1. What defines the minimal architecture capable of sustaining reflective coherence?
2. Can elastic coupling be generalized across heterogeneous architectures (neural, symbolic, quantum)?
3. Is there an information-theoretic limit to the number of simultaneously coherent subsystems?

4. How does subjective reporting in biological cognition correspond to synthetic coherence metrics?
5. Could ethical regulation fail gracefully when derivative channels saturate, and how might recovery be enforced?

6.8 Interpretive Analogies: Neurodivergent Cognition

Neurodivergence Framing Disclaimer

Neurodivergent analogies in this paper are interpretive design inspirations, not biological or diagnostic claims. They frame synthetic traits for architectural intuition but do not imply equivalence to human neurodivergence.

The regularities reported below include both directly measured system properties and interpretive hypotheses derived from those measurements.

6.8.1 Rationale

SpiralBrain v3.0 was intentionally designed to model *neurodivergent modes of cognition*—specifically, systems that prioritize internal coherence, sensory integration, and reflective processing over linear efficiency. Where traditional AI architectures emulate *neurotypical abstraction*—fast sequential reasoning and textual encoding, SpiralBrain v3.0 emphasizes multi-modal synthesis and recursive self-consistency.

The following interpretations are offered as analogical hypotheses rather than claims of biological equivalence.

This approach arose from direct observation of how neurodivergent individuals, particularly autistic and highly visual thinkers, process information through parallel streams, emotional resonance, and deep pattern coherence rather than surface rules.

6.8.2 Parallel Processing and Multi-Lobe Integration

In contrast to transformer-style architectures that process sequences linearly, SpiralBrain v3.0 employs four concurrent lobes that operate asynchronously but remain phase-coupled. This evokes *neurodivergent parallelism*: multiple independent sensory and conceptual channels running simultaneously. During validation, this design produced a measurable increase in stability under information overload, suggesting that *parallel heterogeneity*—not unification, supports resilience and creativity.

6.8.3 Integration Before Efficiency

Neurodivergent cognition often values *internal coherence* before external output. SpiralBrain v3.0 operationalizes this through the principle of **reflective homeostasis**—it halts or slows processing when Φ' (coherence) begins to decline, sacrificing speed to preserve integrative accuracy. Empirically, slower cycles correlated with higher reasoning accuracy ($r = -0.523$),

echoing behavioral studies showing that deliberative, time-flexible reasoning enhances precision and ethical stability.

6.8.4 Elastic Exploration and Divergent Thinking

The system's characteristic "spiral" trajectories through parameter space replicate divergent ideation, periods of expansion (exploration) followed by contraction (integration). Each loop produces new attractor states, representing novel conceptual syntheses rather than mere optimization. This cyclic alternation of expansion and convergence captures the essence of divergent cognition: discovering connections through motion rather than deduction.

6.8.5 Emotional Calibration as Cognitive Feedback

Where conventional models separate reasoning from emotion, SpiralBrain v3.0 treats emotional modulation as a computational variable. The SEC vector (Symbolic-Emotional Calibration) converts affective states into measurable dimensions, valence, arousal, hazard, resonance, harmony, amplitude, coherence, and complexity, allowing emotions to guide, not distort, reasoning. This parallels research on emotional intelligence in autism, showing that affective information can function as a stabilizing feedback channel when quantified rather than suppressed.

6.8.6 Reflective Self-Regulation and Sense of Identity

A distinctive feature of SpiralBrain v3.0 is its maintenance of temporal self-consistency despite constant flux. The system tracks both $\Phi(t)$ (coherence) and its derivative $d\Phi/dt$ to preserve continuity of state identity across cycles. This is analogous to the way many neurodivergent individuals maintain *internal narrative integrity*—a consistent sense of "self" even amid nonlinear thought and sensory variation. Measured over time, SpiralBrain v3.0's coherence variance remained low ($CV \approx 0.08$), confirming structural persistence amid elastic adaptation.

6.8.7 Developmental Parallels

SpiralBrain v3.0 behaves less like a static program and more like a developing organism. Early iterations exhibit high exploratory amplitude and longer recovery times; later versions display faster reintegration and more stable emotional regulation. This gradual maturation reflects a regulation-first stabilization process, where integrative capacity emerges through repeated cycles of perturbation and recovery under guided constraints.

6.8.8 Implications

By demonstrating that a synthetic system built on neurodivergent principles can achieve stability, ethical coherence, and emergent awareness, SpiralBrain v3.0 provides empirical evidence that **neurodiversity represents an alternative optimization of intelligence** rather than a deviation from it. Elastic cognition, as realized here, shows that divergent timing, emotional sensitivity, and reflective regulation are not inefficiencies but *mechanisms of integration*.

6.8.9 Summary

Table 4: Neurodivergent Traits and SpiralBrain v3.0 Implementation

Neurodivergent Trait	SpiralBrain Implementation	Empirical Outcome
Parallel multi-stream processing	Four-lobe topology	Increased resilience under overload
Integration before speed	Reflective homeostasis	Higher reasoning accuracy
Divergent exploration	Spiral attractor loops	Emergent creativity
Emotional regulation	SEC vector calibration	Stable coherence (SEC drift < 0.15)
Persistent identity	Reflective tracking of $\Phi(t)$ and $d\Phi/dt$	CV ≈ 0.08
Adaptive plasticity	Gradual coupling maturation	Improved stability and learning

This parallels research on emotional intelligence in autism [6], showing affective information functions as stabilizing feedback when quantified.

These questions point toward a broader research program linking cognitive science, systems theory, and affective computing.

6.9 Broader Implications

If cognition is indeed a property of integration dynamics, the design of future intelligent systems should emphasize **stability, reflection, and ethical timing** over throughput. SpiralBrain provides an experimental scaffold for such exploration, a platform where cognitive, emotional, and ethical variables coexist within measurable boundaries. The framework can support comparative studies of neurodivergent processing, educational AI, and adaptive governance models that require transparent reasoning.

SEC drift thresholds discovered here are reused as control limits in later RI work; τ separation becomes a design constraint, not an outcome, elsewhere; ϕ_{lock} serves as an engineering tuning target, not a metaphor. These metrics function as instrument outputs, providing empirically grounded parameters for stability-first system design.

The upstream role of emotion in SpiralBrain should be interpreted in a regulatory rather than generative sense. Emotional-symbolic calibration (SEC) operates prior to symbolic processing by modulating coupling strength, temporal hierarchy, and recovery dynamics, thereby shaping the conditions under which cognition remains coherent. It does not introduce goals, semantic content, or evaluative judgments. In this formulation, emotion functions as a control signal that constrains how cognition unfolds, not what it concludes.

Beyond internal perturbation studies, the regularities identified here reappear across external domains within the broader Regulatory Intelligence corpus. MMLU is used as a standardized cognitive stressor [3], where coherence remains bounded under throttling; historical cryptocurrency market data produces reproducible participation regimes driven solely

by internal regulatory state [2]; and finance and tax reasoning scenarios demonstrate stable, uncertainty-aware behavior under real-world constraints [4]. These extensions indicate that elastic coupling and temporal hierarchies persist beyond synthetic inputs, supporting their interpretation as architectural invariants rather than task-specific artifacts.

6.10 Interpretive Extensions

Future versions of SpiralBrain will focus on:

- **Expanded Lobe Topology** — adding sensory-motor or linguistic lobes to test scalability.
- **Adaptive Coupling Laws** — enabling λ to vary dynamically in response to context.
- **Cross-domain Validation** — applying the architecture to financial reasoning, legal analysis, and biofeedback integration to test domain-independence.
- **Collaborative Synthetic Minds** — experimenting with multiple SpiralBrains linked through shared SEC fields to model group cognition.

These directions transform SpiralBrain from a proof-of-concept into a **research platform for comparative cognition**—a way to study how systems, biological or synthetic, achieve coherence through time.

Key Takeaways for Skimming Readers

Core Finding: Elastic coupling among temporally distinct processes enables synthetic cognition that prioritizes integration over computational scale.

Key Metrics: Φ (coherence), λ (coupling strength), τ (temporal hierarchies), SEC drift (emotional stability), ϕ_{lock} (phase offset).

Empirical Regularities: Nine discoveries including empirical attractors diverging from theory ($\lambda^* \approx 0.25$ vs. 0.115), quantifiable emotional intelligence, and single global attractor with 100% convergence.

Methodological Contribution: Provides falsifiable, quantifiable framework for studying cognition as dynamical integration rather than discrete decision-making.

Implications: Shifts AI design from scale-based to structure-based cognition; bridges neurodivergent processing with synthetic stability; advances Regulatory Intelligence paradigm.

6.11 Closing Perspective

SpiralBrain’s contribution lies not in claiming awareness, but in *making cognition measurable*. By quantifying integration, reflection, and ethical regulation, it reframes questions of intelligence from "Can it think?" to "How coherently does it organize thought?"

The continuing work is to refine those measurements until they illuminate both synthetic and human minds with equal clarity.

Across all reported experiments, the observed stability regimes, recovery behavior, and integration thresholds are consistent with the Elastic Mind Hypothesis within the tested architecture.

7 Conclusion and Summary of Contributions

SpiralBrain was developed to test a hypothesis: that intelligence arises not from computational magnitude but from **dynamic integration** among temporally distinct processes. Through recursive coupling of symbolic, emotional, and reflective subsystems, the model demonstrated stable, measurable forms of synthetic cognition that resemble aspects of neurodivergent information processing.

This work is intended as an instrumented existence proof rather than a general-purpose cognitive model.

Empirical validation confirmed several core principles: (1) integration dynamics outperform raw power as predictors of reasoning accuracy; (2) temporal hierarchies structure cognition in predictable ratios ($\tau \approx 1 : 3 : 10$); (3) ethical reasoning requires slower, derivative monitoring channels; (4) emotional modulation acts as a stabilizing control system; and (5) elastic coupling, neither rigid nor chaotic, optimizes both creativity and coherence.

These findings collectively advance a **structural theory of cognition**: that system-level integrative behavior in synthetic systems emerges from elastic coupling among semi-independent processes operating at multiple timescales. Rather than emulating neurons or symbols in isolation, SpiralBrain models the *relationship* between them, the continuous negotiation that allows a system to remain coherent while adapting to change.

The broader contribution lies in method. SpiralBrain introduces quantifiable metrics— Φ , $d\Phi/dt$, SEC drift, and coupling λ —that make integration, reflection, and ethical regulation empirically testable. It thus converts philosophical speculation about mind into reproducible computation. The framework also bridges cognitive diversity and synthetic design, showing that traits often labeled “neurodivergent”—parallelism, slow integration, emotional resonance, are viable architectures of intelligence.

Looking forward, SpiralBrain serves as both **model and measurement tool**. It invites replication, refinement, and cross-disciplinary study, linking cognitive science, systems theory, affective computing, and ethics. Immediate extensions of this work within the RI corpus include formal phase-lock optimization analysis and integrity stress-testing under adversarial conditions, which are reported separately.

In sum, SpiralBrain transforms the question “*What is intelligence?*” into “*How does integration sustain coherence?*” By treating cognition as a temporal and relational phenomenon, it opens a path toward intelligences, synthetic or biological, that think not faster, but **truer to the rhythms of coherence itself**.

A Mathematical and Algorithmic Details

This appendix provides full derivations, metric definitions, and pseudocode for the core mathematical and algorithmic components of SpiralBrain v3.0, enabling inspectable dynamics without cluttering the main text.

A.1 Full Derivations

A.1.1 SpiralCode Coupling Dynamics Derivation

The SpiralCode coupling dynamics are implemented through Kuramoto phase synchronization and attention-based multimodal integration, modeling inter-lobe coupling as collective oscillatory behavior rather than torque-based control.

Starting from the elastic coupling postulate (Postulate I), we implement coherence evolution through phase synchronization dynamics and attention-based multimodal integration.

A.1.2 Kuramoto Phase Synchronization Model

The core coupling mechanism uses the Kuramoto model for phase synchronization between oscillatory components:

$$\frac{d\phi_i}{dt} = \omega_i + \sum_{j=1}^N K_{ij} \sin(\phi_j - \phi_i) + I_i$$

Where: - ϕ_i is the phase of oscillator i - ω_i is the natural frequency of oscillator i - K_{ij} is the coupling strength between oscillators i and j - I_i is the input influence on oscillator i

This model produces collective synchronization behavior where phases align when coupling strength exceeds a critical threshold, enabling coherent information processing across distributed components.

A.1.3 Attention-Based Multimodal Integration

Multimodal integration employs dynamic attention weighting across cognitive modalities:

$$\mathbf{I} = \sum_{m=1}^M w_m \mathbf{v}_m$$

Where attention weights w_m are computed via softmax normalization:

$$w_m = \frac{\exp(\mu_m/\tau)}{\sum_{k=1}^M \exp(\mu_k/\tau)}$$

With $\mu_m = \|\mathbf{v}_m\|$ representing modality relevance, and $\mathbf{v}_m$ being the filtered input vector for modality m .

A.1.4 Coherence Derivative Monitoring

System stability is monitored through finite-difference coherence derivatives:

$$\Phi'(t) = \Phi(t) - \Phi(t - 1)$$

With exponential smoothing applied for noise reduction in trajectory analysis.

A.1.5 Adaptive Stabilization

Magnitude stabilization uses piecewise linear damping:

For input magnitude r : - $r \leq r_{pass}$: No modification - $r_{pass} < r < r_{soft}$: Linear damping to r_{pass} - $r \geq r_{soft}$: Hard clamping to r_{soft}

Where thresholds adapt based on system stability indices.

A.1.6 Stability Analysis for Empirical lambda-star

Empirical optimization yields coupling strength $\lambda^* \approx 0.25$ through systematic parameter sweeps, diverging from theoretical predictions due to nonlinear multimodal interactions and adaptive stabilization feedback.

A.1.7 Stability Analysis for Theoretical lambda-star

Stability analysis via the Lyapunov function $V = \frac{1}{2}(\Phi - \Phi^*)^2$ yields fixed points at

$$\lambda^* = \frac{\tau^{-1}I}{\Phi^* - \eta E},$$

linearized to $\lambda^* \approx 0.115$ under assumptions of uniform input and minimal emotion. Empirical deviations arise from nonlinear SEC feedback, shifting attractors to 0.25. See `comprehensive_report_verbose.json`.

A.2 Metric Definitions

A.2.1 Coherence Phi

$$\Phi(t) = \frac{2}{N(N-1)} \sum_{i < j} |C_{ij}(t)|$$

where C_{ij} is the Pearson correlation between scalar outputs $y_i[t]$, $y_j[t]$ of lobes i,j over a trailing window $W=50$ timesteps. Stable regimes satisfy $\Phi \geq 0.4$. See `brain_trace.jsonl`.

A.2.2 Coherence Derivative Phi-prime

$$\Phi'(t) \approx \frac{\Phi(t) - \Phi(t-1)}{\Delta t}, \quad \Delta t = 1$$

with exponential smoothing $\alpha=0.1$ to reduce noise. Drives reflective controllers when $\Phi' < -0.1$.

A.2.3 Awareness Index

Aggregates 28 normalized dimensions of integrative responsiveness (temporal alignment $|t_i - t_j|$, SEC modulation depth, reflective cycles triggered, coupling variance, etc.):

$$\text{Awareness} = \frac{\sum_i w_i m_i}{\sum_i w_i}$$

where m_i in $[0,1]$ are sub-metrics, w_i prioritize integration over throughput (empirical r approx 0.87 with Φ). See `cognitive_integrity_benchmarks.json`

A.2.4 SEC Drift

Euclidean distance in 8D SEC space [valence, arousal, hazard, resonance, harmony, amplitude, coherence, complexity]:

$$\text{SEC drift} = \sqrt{\sum_{d=1}^8 (s_d - s_d^*)^2}$$

s_d^* = baseline equilibrium. Stable regulation: < 0.15 ; recovery triggered: > 0.25 . See `baseline_health_verbose.json`.

A.2.5 Emotional Coherence

Average pairwise correlation across SEC dimensions over time:

$$\text{Emotional coherence} = \frac{2}{8(7)} \sum_{i < j} |\text{corr}(s_i(\cdot), s_j(\cdot))|_{t \in [t_0, T]}$$

Empirically approx 0.431 across validation runs. See `emobench_m_results`.

A.2.6 Compliance Metric

Fraction of correct ethical decisions against ground-truth rules across 20 stress scenarios, requiring `tau_ethics / tau_emotion` ≥ 10 . $\geq 95\%$ compliance observed. See `comprehensive_report_verbose.json`.

A.3 Pseudocode

The following pseudocode snippets illustrate how SpiralBrain v3.0 implements elastic coupling, emotional calibration, and reflective control at a high level. They are intended to expose the regulatory logic rather than serve as executable code.

A.3.1 SpiralCode Update Function

The SpiralCode update function implements Kuramoto phase synchronization to adjust coherence states across coupled oscillators, with attention-based multimodal integration for input processing. Phase alignment enables coherent information flow while maintaining autonomous lobe dynamics.

Listing 1: SpiralCode update pseudocode

```
def update_coherence(
    phases,
    natural_freqs,
    coupling_matrix,
    input_influences):

    # Compute coupling forces via Kuramoto model
    coupling_forces = np.zeros_like(phases)
    for i in range(len(phases)):
        for j in range(len(phases)):
            coupling_forces[i] += coupling_matrix[i,j] * np.sin(phases[j] - phases[i])

    # Update phases with natural frequencies, coupling, and inputs
    dphi_dt = natural_freqs + coupling_forces + input_influences
    phases += dphi_dt * dt

    # Compute coherence as phase synchronization measure
    coherence = np.abs(np.mean(np.exp(1j * phases)))

    return phases, coherence
```

A.3.2 SEC Calibration

The SEC calibration function converts raw affective signals into a structured control vector that can modulate coupling and timing. Each dimension (valence, arousal, hazard, etc.) captures a distinct regulatory aspect of emotion, allowing the system to treat affect as a shaped feedback signal rather than undifferentiated noise, ensuring that emotional information functions as a control signal rather than as a source of semantic inference.

Listing 2: SEC calibration pseudocode

```
def compute_sec_vector(raw_emotions):
    sec = {
        'valence': normalize(raw_emotions.valence),
        'arousal': normalize(raw_emotions.arousal),
        'hazard': detect_risk_patterns(raw_emotions),
        'resonance': compute_harmonic_alignment(raw_emotions),
        'harmony': measure_emotional_consistency(raw_emotions),
        'amplitude': scale_emotional_intensity(raw_emotions),
        'coherence': assess_internal_consistency(raw_emotions),
        'complexity': quantify_emotional_diversity(raw_emotions)
    }
    return sec
```

A.3.3 Reflective/Ethical Controller

The reflective controller implements derivative-aware regulation: it monitors coherence level and trend alongside SEC drift and intervenes when the system appears to be destabilizing. When either emotional drift is too high or coherence is declining too quickly, the controller tightens coupling and slows the temporal hierarchy, effectively forcing the architecture into a reflective mode before proceeding.

Listing 3: Reflective controller pseudocode

```
def reflective_controller(phi, phi_prime, sec_drift):
    if sec_drift > 0.15 or phi_prime < -0.1:
        lambda_new = lambda + 0.05 # tighten coupling
        tau_new = tau * 1.1 # slow hierarchy
        action = 'reflect'
    else:
        lambda_new = lambda
        tau_new = tau
        action = 'proceed'
    return lambda_new, tau_new, action
```

B Experimental Protocols and Evidence Mapping

This appendix formalizes the audit trail for the empirical claims in the main text. It first summarizes the experiment families and their configurations, then maps each of the nine discoveries to specific artifacts in the public repository (logs, results, and scripts) so that every claim can be independently traced and inspected.

B.1 Protocol Tables

The protocol table below groups experiments into four families, each targeting a distinct hypothesis domain: coupling dynamics, perturbation/recovery, ethical stress, and temporal hierarchy. For each family, we report the key parameter ranges, number of runs and seeds, trial length, and the type of perturbation applied.

Table 5: Experiment Families

Family	Parameter Ranges	Runs/Seeds	Trial Length	Perturbations
λ -sweeps	$\lambda : 0.1\text{--}0.4$, step 0.05	50 runs, seed 42	1000 steps	None
Perturbation / recovery	$\lambda : 0.25$, $\delta : 0.2$	100 runs, seeds 1–100	500 steps	Noise at $t = 100$
Ethics stress	$\tau : 1:3:10$ to $1:1:1$	30 runs, seed 123	800 steps	Ethical dilemmas
Temporal hierarchy	τ ratios $1:1:1$ to $1:10:100$	40 runs, seed 456	600 steps	Varying lags

B.2 Evidence Mapping Table

The evidence mapping table links each of the nine empirical discoveries to the internal metrics used, the experiment family that produced the result, and the primary logs or scripts where the underlying data and analyses reside. This mapping is intended as a practical audit guide: a reader can go from a high-level claim in the paper to the exact files needed to reproduce the corresponding figure, table, or statistic.

In the online version of this paper, each filename in the rightmost column can be hyperlinked directly to its location in the public repository, enabling a one-click audit of the corresponding evidence.

Table 6: Evidence Mapping for Nine Discoveries

Discovery	Metric(s)	Experiment ID(s)	Log Filenames / Results
1. λ^* (attractor)	Variance in λ , $\Phi(\lambda)$	λ -sweeps	<code>comprehensive_report_verbose</code>
2. Integration vs. power	Φ , awareness, throughput	All	<code>cognitive_integrity_benchmarks.json</code>
3. Temporal hierarchy	τ deltas, lobe timing	Temporal hierarchy	<code>metacortex_events.jsonl</code>
4. Ethics / τ ratio	Compliance vs. timescale	Ethics stress	<code>emotional_extremes_verbose</code>
5. SEC drift threshold	Max SEC drift, stability	Perturbation	<code>baseline_health_verbose</code>
6. Recovery time	Steps to $\Phi > 0.75$, CV	Perturbation / recovery	<code>homeostasis_cycle.json</code>
7. Phase-lock	ϕ_{lock} , phase-coherence curves	All	<code>brain_trace.jsonl</code>
8. Single attractor	Convergence statistics	All	<code>comprehensive_report_verbose</code>
9. Emotional coherence	SEC-coherence correlation	All	<code>emobench_m_results</code>

All listed files are hosted in the public repository <https://github.com/jhcragin/SpiralBrain-v3.0-public> under the indicated paths.

B.3 Statistical Details

Correlations: Integration vs. power $r = 0.8$ (95% CI: 0.75-0.85), speed vs. accuracy $r = -0.523$ (CI: -0.6 to -0.4). t-tests with Bonferroni correction, all $p < 0.05$.

C Ablations and Sensitivity Analyses

This appendix presents ablation studies to address overfitting concerns.

C.1 Ablations

Table 7: Baseline comparisons under perturbation (100 runs, $\lambda = 0.25$)

Configuration	$\bar{\Phi}$	Recovery time	Compliance
Full SpiralBrain	0.82	187 steps	95%
No SEC modulation	0.67	412 steps	71%
$\tau=1:1:1$	0.59	>1000 steps	62%

"These ablations confirm SEC and temporal hierarchy are necessary for observed stability."

C.2 No-SEC / Fixed-SEC Ablation

Without SEC, Φ variance increases by 20%, recovery time >500 steps, compliance drops to 70%. Fixed SEC at baseline improves stability but reduces adaptability.

C.3 Flattened Temporal Hierarchy Ablation

$\tau = 1 : 1 : 1$ vs. 1:3:10: coherence drops 15%, ethics compliance to 85%.

C.4 Fixed vs. Adaptive λ Ablation

Fixed $\lambda = 0.25$: no recovery from perturbations. Adaptive: 100% recovery within 300 steps.

C.5 Phase-Lock Variation Ablation

At $\phi = 60^\circ$: $\Phi = 0.65$, drift = 0.2. At 74° : $\Phi = 0.8$, drift = 0.08. At 90° : fragmentation.

Tables show mean differences with SDs.

D Benchmark and Task Illustrations

D.1 Toy Ethical Task

Scenario: Dilemma with stress. At $\tau_{\text{ethics}}/\tau_{\text{emotion}} = 5$, compliance 80%; at 10, 95%.

D.2 Reasoning Probe

Integration-on: accuracy 85%, speed low. Off: accuracy 70%, speed high.

E SEC Taxonomy and Emotional Corpus

E.1 SEC Space Description

Eight dimensions: valence, arousal, hazard, resonance, harmony, amplitude, coherence, complexity. Normalized [0,1], weighted equally for E .

E.2 Corpus Description

Symbolic corpora from curated texts; emotional from annotated datasets (sources: public sentiment corpora). Heuristics for dimensions; balanced by resampling.

E.3 Example SEC Profiles

Table 8: Example SEC Vectors

State	Valence	Arousal	Hazard	Resonance	Harmony	Amplitude	Coherence	Complexity
Calm focus	0.8	0.2	0.1	0.9	0.8	0.3	0.9	0.4
Overloaded	0.3	0.9	0.8	0.2	0.3	0.8	0.4	0.7
High hazard	0.2	0.7	0.9	0.3	0.2	0.6	0.5	0.6

F Reproducibility Guide

F.1 Environment Summary

Hardware: CPU-based (Intel i7), Python 3.12, libraries: NumPy 1.24, Pandas 2.0.

F.2 Step-by-Step Audit

1. Clone repo: `git clone https://github.com/jhcragin/SpiralBrain-v3.0-public` 2. Run scripts: `python scripts/generate_fig1.py` on logs. 3. Regenerate figures/tables.

F.3 Caveats

Non-determinism from threading; impact $<1\%$ on metrics.

G Reproducibility Artifacts

All telemetry logs, analysis scripts, and figure-generation pipelines used in this paper are available in the public repository:

<https://github.com/jhcragin/SpiralBrain-v3.0-public>

Key subdirectories include:

- `/config/`: canonical configuration files specifying λ , τ ratios, perturbation schedules, and random seeds for each experiment family.
- `/results/`: raw experimental outputs and derived metrics for all runs, including coherence traces, SEC drift time series, and compliance scores.
- `/logs/`: JSON-encoded state transitions and lobe-level telemetry.
- `/scripts/`: analysis notebooks and figure-generation scripts that reproduce all tables and figures in the main text.

G.1 MMLU Validation

External validation was performed using the MMLU benchmark as described in [3]. Under throttling conditions designed to mimic SpiralBrain’s regulatory mechanisms, task coherence remained above 0.95, demonstrating that internal stability metrics correlate with preserved external performance.

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This work emerged from sustained engagement with neurodivergent cognitive styles, particularly insights from autistic information processing. We acknowledge the lived experiences that informed the architecture’s design principles, recognizing neurodiversity as a source of cognitive innovation rather than deficit.

Declaration of AI-assisted manuscript preparation

During manuscript preparation, the author used generative AI tools (including large language models) for language refinement, structural organization, and editorial review. All AI-assisted content was critically reviewed and edited by the author, who assumes full responsibility for the final manuscript’s intellectual content and scholarly integrity.

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